

1.5 The functions of mechanical devices used to produce different sorts of movements, including the changing of magnitude and the direction of forces	The performance, principles, applications and the influence on the design of products of the following.
	1.5.1 Types of movement: a linear b reciprocation c rotary d oscillation.
	1.5.2 Classification of levers: a class 1, 2 and 3 b calculations related to mechanical advantage (MA), velocity ratio (VR), load, effort and efficiency.
	1.5.3 Linkages: a bell crank b reverse motion linkages.
	1.5.4 Cams: a pear shaped b eccentric (circular) c drop (snail).
	1.5.5 Followers: a roller b knife c flat followers.
	1.5.6 Pulleys and belts: a V-belt b velocity ratio (VR) c input and output speeds.

CORE

Key idea	What students need to learn:
	1.5.7 Cranks and sliders.
	1.5.8 Gear types: - a simple and compound gear train - b idler gear - c revolutions per minute (RPM) calculations - d bevel gears - e rack and pinion.